

AYE
TECH HUB

• PROTASTRUCTURE • LESSON 04

Grid and Level Setup

Stories · Axes · Materials · Analysis Settings · First Model

Lesson 04 of 30 | ProtaStructure Program

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- Creating stories and setting floor elevations for a real building
- Drawing the structural axis grid in plan view — step by step
- Setting concrete grade, rebar grade and cover depth correctly
- Configuring the analysis model: code, load combos, seismic data
- Pre-modelling verification checklist before placing the first column

The Project: A 5-Storey RC Office Building

In this lesson we build the complete structural framework in ProtaStructure for a real project: a 5-storey reinforced concrete office building. This is the work done **before placing a single beam or column**. By the end you will have: stories configured with correct elevations, a structural grid drawn, materials set, and the analysis model configured and ready for element input.

Step 1 — Creating Stories in the Building Tree

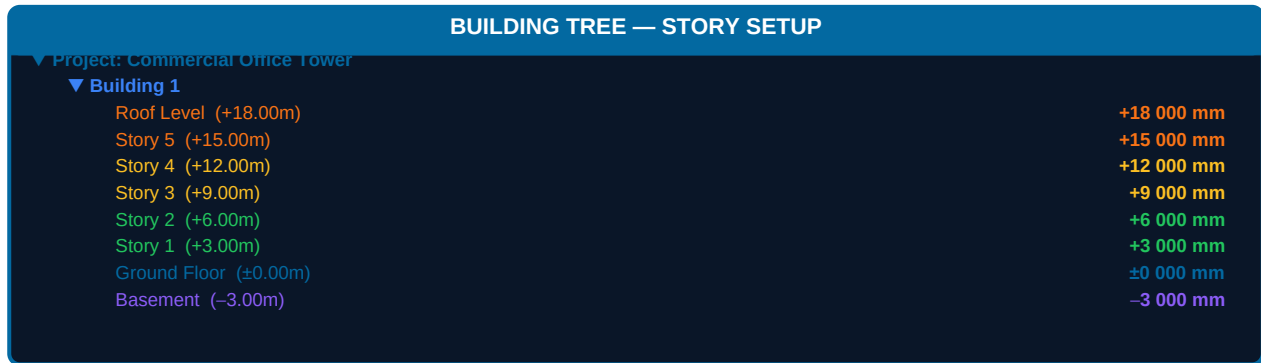


Figure 1 — ProtaStructure Building Tree: 9 Stories from Basement to Roof

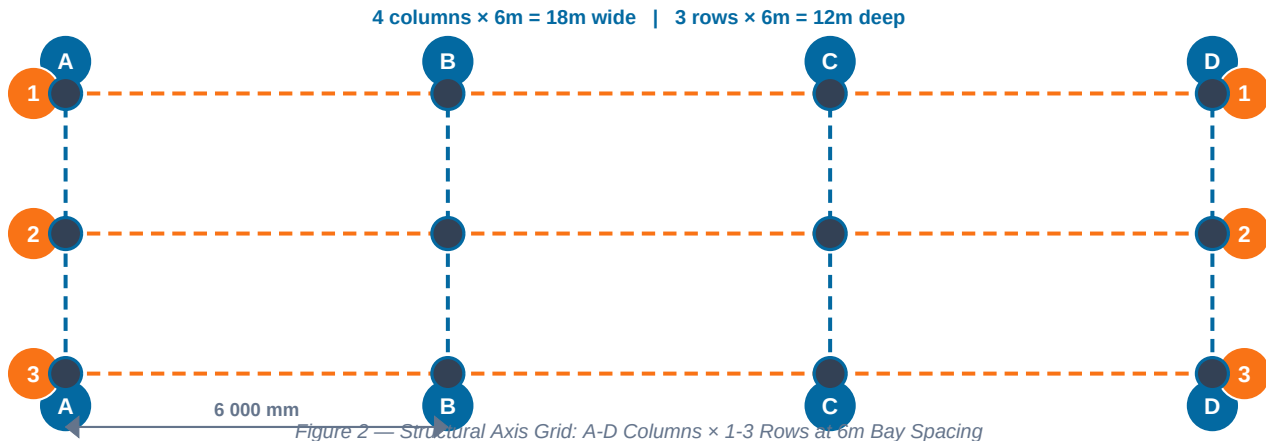
Story	Name in ProtaStructure	Floor Height (mm)	Notes
Story 1	Ground Floor	3 000	Datum — all elevations measured from here (0mm)
Story 2	First Floor	3 000	Typical office floor, 3000mm clear height
Story 3	Second Floor	3 000	Same as first — use "Add Same Story" button
Story 4	Third Floor	3 000	Same as first
Story 5	Fourth Floor	3 000	Same as first
Story 6	Roof Level	3 000	Structural roof slab level — no habitable space
Story 0	Basement	3 000	Below ground — set as negative elevation (–3000mm)

Action	ProtaStructure Method	Key Setting
Create first story	Building Tree → right-click "Building 1" → Add Story Type name: "Ground Floor"	Floor height: 3000mm. This is the bottom of the slab, not top.
Add stories above	Right-click "Ground Floor" → Add Story Above Or use "Add Same Story" button (faster)	ProtaStructure auto-numbers and stacks stories. Rename each.
Add basement below	Right-click "Ground Floor" → Add Story Below Name: "Basement"	ProtaStructure automatically assigns negative elevation (–3000mm).
Set storey height	Double-click story name → Properties → Floor Height field Type 3000 (mm)	Floor height = distance from this floor to the one above. Not room clear height.
Verify all elevations	Look at Building Tree — each story shows its absolute elevation Ground: 0, First: +3000, Second: +6000, etc.	Roof Level should show +15000mm if 5 floors of 3000mm each.

Story height vs clear height: In ProtaStructure, the story height is measured from the top of one slab to the top of the slab above. If your slabs are 200mm thick and you need 2800mm clear height, the story height = 2800 + 200 = 3000mm. Always confirm with the architect's section before setting story heights.

Step 2 — Drawing the Structural Grid

With stories set, switch to the plan view of Ground Floor and draw the structural axis grid. Our building has a regular 4×3 grid at 6m bays — 4 column lines across (A-D) and 3 column lines deep (1-3). This gives 12 column positions and 6 structural bays.



Step	Action in ProtaStructure	Result
1	Activate Ground Floor in Building Tree (double-click) Switch to 2D plan view (toolbar → "2D" button)	Plan view shows empty floor. Grid will appear here.
2	Input menu → Axes → Define Axis System Or click the grid icon in toolbar	Axis definition dialog opens. Define X-axes (A,B,C,D) and Y-axes (1,2,3).
3	X-axis (vertical grid lines A-D): Starting X: 0. Spacing: 6000, 6000, 6000 (3 gaps for 4 lines)	Axis lines A, B, C, D drawn at x=0, 6000, 12000, 18000mm
4	Y-axis (horizontal grid lines 1-3): Starting Y: 0. Spacing: 6000, 6000 (2 gaps for 3 lines)	Axis lines 1, 2, 3 drawn at y=0, 6000, 12000mm
5	Click OK/Apply — axes appear in plan view Verify axes by counting lines and checking spacing labels	Grid is now set for the entire building — all floors share the same grid
6	Propagate axes to all stories: Input → Axes → Propagate to All Stories	Grid lines appear on every floor plan automatically

Grid naming in ProtaStructure: You can name axes freely. Match the names exactly to the structural engineer's drawings (e.g. if the SE uses "A1" and "B1" instead of "A" and "B", use the same). Grid names appear on all plans, sections and schedules. Renaming grids after modelling is very disruptive.

Step 3 — Materials, Analysis Settings and Pre-Modelling Checks

Before placing any structural elements, set the material grades and configure the analysis model. Getting these wrong means the design results are based on incorrect parameters — the most dangerous error in structural software.

CONCRETE GRADE

C25/30 → fck = 25 MPa (Ethiopia/Africa common)
C30/37 → fck = 30 MPa (standard commercial)
C35/45 → fck = 35 MPa (columns under high load)
C40/50 → fck = 40 MPa (transfer slabs, PT)

ProtaStructure: Define → Materials
→ Concrete → select or create grade
Ec = 22(fck/10)^0.3 GPa (EC2)

REBAR GRADE

B500B (EC2) → fyk = 500 MPa
Grade 60 (ACI) → fy = 414 MPa
HYSD 500 → fy = 500 MPa (India/Africa)

Most common: B500B for EC2 design
Higher yield → less steel needed

ProtaStructure: Define → Materials
→ Rebar → set yield strength

COVER DEPTH

Exposure Class XC1 (interior): 15–20mm
Exposure Class XC2 (external): 25–30mm
Exposure Class XS1 (marine): 40–45mm

Cover = distance from bar surface to concrete face (not centreline)

ProtaStructure: Define → Cover
Set per element type (beam/column/slab)

Figure 3 — Material Grades: Concrete / Rebar / Cover Depth

Analysis Model Configuration

Setting	Where to Set	Typical Value / Guidance
Design code	Define → Project Settings → Design Code	Eurocode 2 (EN 1992) for Africa/Europe; ACI 318 for USA/Middle East
Load combinations	Define → Load Cases & Combinations → Review and confirm	EC2: 1.35G+1.5Q; 1.35G+1.05Q+1.5W. Confirm all required combinations present.
Seismic zone / data	Define → Seismic Settings → input Ss, S1, soil class, importance factor	Use local seismic hazard map. For Ethiopia: PGA 0.1–0.25g depending on zone.
Slab self-weight	Define → Materials → Concrete unit weight	25 kN/m³ for reinforced concrete (default — verify)
Mesh size for slabs	Analysis → FEM Settings → Element Size	500–750mm for standard slabs; smaller = more accurate but slower
Analysis type	Analysis → Analysis Type → Linear static vs Response spectrum	Use response spectrum analysis if seismic governs (high-seismicity zones)

Pre-Modelling Checklist

Check	How to Verify	If Wrong, Fix By
All stories created with correct heights	Building Tree → check all story names and elevations	Double-click story → Properties → correct floor height
Grid axes drawn and named correctly	Open Ground Floor plan → count grid lines, verify labels	Input → Axes → Edit Axis System to modify
Grid propagated to all stories	Open each floor plan — axes visible on all	Input → Axes → Propagate to All Stories
Concrete grade set correctly	Define → Materials → Concrete → confirm fck value	Edit material definition — apply to project
Rebar grade set correctly	Define → Materials → Rebar → confirm fy value	Edit material definition — apply to project
Cover depth set per element type	Define → Cover → check beam, column, slab covers	Edit cover settings per element type
Design code confirmed	Define → Project Settings → Design Code	Select correct code before placing any elements

Load combinations reviewed	Define → Load Combinations → review all listed combos	Add or modify combinations as required by project spec
Seismic data entered	Define → Seismic → check all seismic parameters populated	Input seismic zone data from local hazard map or code

What comes next — PROTA-005: Beam Design — inputting beams onto the structural grid, setting beam cross-sections, running the first analysis, reviewing bending moment diagrams, and checking the initial beam design results.